

Strategic Under-Specification: A Stackelberg Game Between User and Assistant

Pranjal Agarwal
BITS Pilani, Hyderabad Campus

Abstract

Users of coding assistants routinely under-specify their intent, and the assistant’s policy for handling missing information—guess silently, or ask a clarifying question—is itself a strategic choice that users respond to. We formalize this as a Stackelberg game: the assistant commits to an ask/guess policy, and a population of users with privately known, heterogeneous specification costs best-respond. We first solve a decision-theoretic benchmark in which the policy is exogenous. Here, improving the assistant’s raw guessing accuracy alone cannot make users worse off, even though it induces them to specify less—an immediate envelope-theorem consequence of letting a single agent optimize freely against an improved technology, but one worth stating explicitly because it is the benchmark against which the paper’s actual moral-hazard channel (a gap between believed and true policy) should be measured; naive “better guessing causes worse specs and worse outcomes” stories conflate the two. We then close the loop and solve for the assistant’s own optimal policy under two regimes: a *pooling* policy restricted to a single population-wide ask rate, and a *screening* policy that can condition on the user’s declared specification. We show that an unbiased (welfare-aligned) assistant achieves full efficiency under screening despite users’ private information—incentive compatibility is free when principal and agent share an objective—so all screening-regime inefficiency is attributable to misalignment between the assistant’s effective objective and true user welfare, not to private information per se. Pooling, by contrast, loses efficiency from restricted instruments even when unbiased. We characterize both equilibria in closed or near-closed form under a tractable linear-risk specification, derive testable comparative statics, evaluate them via numerical and LLM-based simulation experiments, and outline a proposed human-study design for future work.

1 Introduction

A user typing a coding request faces a choice: spend effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training) with some implicit model of how users will respond. Section 2 treats the assistant’s policy as exogenous and solves the user’s problem against it—a one-sided optimization, useful as a benchmark but not yet a game. Sections 4–7 close the loop: the assistant is a Stackelberg leader that commits to a policy anticipating the population’s best response, and we solve for that policy under two different instrument restrictions, then compare them.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection models (Myerson, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening and costly verification (Townsend, 1979), where information revelation is governed by operational friction rather than cash transfers.

Costly communication and strategic disclosure. The user’s specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user’s rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level interaction connects to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016; von Stackelberg, 1934), where an autonomous assistant acts to maximize human utility under incomplete information. However, while assistance games formalize the assistant’s uncertainty over the human’s underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). Instead, the informational asymmetry resides entirely in the user’s private *specification cost* κ . The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. To our knowledge, our model provides the first contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification and interactive NLP. A parallel empirical literature in natural language processing and information retrieval investigates algorithms for generating and ranking clarifying questions in conversational search and dialog (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020). These works optimize static clarification decisions or train heuristic dialog policies via reinforcement learning against fixed datasets. However, they uniformly treat user inputs as exogenous test instances rather than the output of an optimizing agent who strategically anticipates the clarification policy. Our framework provides the contract-theoretic microfoundation that this engineering literature lacks, formally endogenizing prompt specification effort as an equilibrium best response to the assistant’s asking policy.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in$

$\{0, \dots, k\}$ and truthfully reveals m attributes.¹ Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0, 1)$. We model the assistant's policy at this stage as a uniform ask rate $a \in [0, 1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a, g) = a + (1 - a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user's payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a(k - m)\kappa,$$

with $V > 0$ the value of success and $\kappa > 0$ the user's cost of answering one clarifying question. We treat m as continuous on $[0, k]$ for tractability.

2.2 Proposition 1: accuracy alone does not create moral hazard

Proposition 1. *Fix a . Let $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$. Then U^* is weakly increasing in g .*

Proof. By the envelope theorem, at the optimum $m^*(g)$, $dU^*/dg = \partial U / \partial g|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0$. The envelope theorem lets us ignore dm^*/dg : because m^* is chosen optimally, its first-order response to g has no first-order effect on U^* . $\square$

Assumption 1 (Regularity). *In the relevant range of m , $(k - m) \ln q(a, g) \geq -1$.*

Remark 1 (Interpreting Assumption 1, and Monotonicity Survival). *The condition has no closed-form economic reading on its own; it is a sufficient condition for the single-crossing property used in Corollary 1 and Proposition 2 via Topkis's monotone comparative statics theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994), not a claim about primitives directly. The direction is pinned down by the cross-partial itself: for fixed a ,*

$$\frac{\partial^2 U}{\partial m \partial g} = -V(1 - a)q^{k-m-1}[(k - m) \ln q + 1],$$

so U has decreasing differences in (m, g) — the condition Topkis's theorem (Topkis, 1978, 1998) needs to conclude $m^(g)$ is weakly decreasing — exactly when $(k - m) \ln q + 1 \geq 0$, i.e., when Assumption 1 holds as stated here. If the inequality were reversed, U would exhibit increasing differences and hence $m^*(g)$ would be locally increasing (e.g., at $V=10, k=5, a=0, c(m)=m^2/2$, m^* rises from ≈ 0.26 at $g=0.5$ to ≈ 0.67 at $g=0.9$ before turning over and falling as $g \rightarrow 1$). Loosely, since $\ln q \approx -(1 - q)$ for q near 1, the condition is close to $(k - m)(1 - q) \leq 1$: the expected number of wrong attributes under the current policy is at most about one. A numerical grid evaluation over $k \in [5, 15]$, $q \in [0.70, 0.95]$, and $\kappa \in [0.20, 2.00]$ (726 grid points, quadratic cost $c(m) = \kappa m^2/2$, evaluated at each point's own optimizing m^*) finds that under the benchmark simulation valuation ($V = 100$), the condition holds on 89.9% (653/726) of this parameter space. In lower-valuation regimes where specification effort is lower ($V = 10$, as in the numerical example above), the condition holds on roughly 34% (and 31.7% at $V = 8$); the exact share is sensitive to the scaling of $c(m)$ relative to V . Crucially, across the tested parameter grid (2,552 parameter intervals), a comprehensive finite-difference evaluation confirms that monotonicity of specification*

¹We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

$(m^*(g)$ weakly decreasing in g) holds without exception across all 2,162 intervals where Assumption 1 is satisfied (100% survival, zero violations). Outside the sufficient condition $((k - m) \ln q < -1$, 390 intervals), the cross-partial turns strictly positive and monotonicity breaks in 96.7% of tested intervals (377/390), confirming that the regularity condition is empirically tight.

Corollary 1. *Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g) ; Topkis’s theorem).*

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U / \partial m \partial g \leq 0$ on the relevant range, which is decreasing differences in (m, g) ; Topkis’s theorem then gives $m^*(g)$ weakly decreasing. $\square$

So, holding the assistant’s policy fixed, users specify less as the assistant gets better at guessing, yet are never worse off for it. We flag explicitly that Proposition 1 is a low bar: it is a direct application of the envelope theorem to a single agent optimizing freely against a technology parameter that enters its payoff with non-negative derivative, and would hold in essentially any model of this form. Its purpose here is not to be a surprising finding but to be the correct benchmark against which any claimed moral hazard should be measured, so that Proposition 2 below can be read as identifying the actual, non-trivial source of harm. It isolates the question this paper is really about: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates this response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant’s ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (a reviewer, downstream production systems) bears part of the cost of a wrong guess, the user’s private (κ, V) understate the social values, and the user under-specifies relative to the social optimum at any fixed g —worsening, by Corollary 1, as g rises. Standard; not our focus.

3.2 Miscalibrated belief updating under silent failure

If wrong guesses are not always detected, and detection probability $d(m)$ is increasing in m , users infer g from experience rather than observing it, and under-specifying users are also less likely to catch their own errors—an upward-biased $\hat{g}$ that reinforces under-specification. This is a dynamic learning story rather than a static equilibrium result; we return to it in the empirical design (prediction P5).

3.3 Belief about the assistant’s policy

Let users be *sophisticated* (correctly infer the assistant’s true ask rate a_A) or *naive* (believe the assistant follows the welfare-optimal rate $a^\dagger \geq a_A$).

Proposition 2. *Under Assumption 1, naive users choose $m_{naive}^* \leq m_{soph}^*$ and $U(m_{naive}^*; a_A, g) < U(m_{soph}^*; a_A, g)$: naive users under-specify and are strictly worse off than sophisticated users, holding raw guessing accuracy g fixed throughout.*

Proof. Because $q(a, g) = a + (1 - a)g$ is strictly increasing in a for $g \in (0, 1)$, a naive user believing $a^\dagger \geq a_A$ perceives a higher resolution accuracy $q(a^\dagger, g) \geq q(a_A, g)$. Under Assumption 1, Corollary 1 proves that $m^*(g)$ is weakly decreasing in accuracy, so $m_{\text{naive}}^* \leq m_{\text{soph}}^*$. Furthermore, $U(m; a_A, g)$ is strictly concave in m with unique maximizer m_{soph}^* , so any choice $m_{\text{naive}}^* \neq m_{\text{soph}}^*$ delivers strictly lower true utility: $U(m_{\text{naive}}^*; a_A, g) < U(m_{\text{soph}}^*; a_A, g)$. $\square$

This is the source of genuine moral hazard identified so far: not raw accuracy, but an *unobserved gap between the user's belief about the assistant's policy and the assistant's true policy*. Section 5 solves for what that true policy a_A actually is when the assistant is itself a strategic, possibly biased, Stackelberg leader—so Proposition 2 can be instantiated with a derived a_A rather than an assumed one.

4 The Assistant's Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m; \kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$. The leader's own objective Π need not equal U : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q) a (k - m),$$

where $\mu_A \in (0, 1]$ discounts the leader's weight on true (downstream) user welfare, c_Q is the user's true cost of answering one clarifying question, and λ_A is the leader's own *perceived* per-question friction cost, which may differ from c_Q (e.g., a training signal that penalizes extra turns more than users actually mind answering a direct question). The leader is *unbiased* iff $\mu_A = 1$ and $\lambda_A = c_Q$, in which case $\Pi_\kappa \equiv U(\cdot; \kappa)$ exactly. The leader solves $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F} [\Pi_\kappa(m^*(\kappa; \pi), \pi)]$, a bilevel program.

Remark 2 (Existence). *For Model I below ($\pi = a \in [0, 1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a (strict concavity of U in m plus the implicit function theorem), so $\mathbb{E}_\kappa[\Pi_\kappa]$ is continuous in a and a maximizer exists on $[0, 1]$ by Weierstrass's theorem. For Model II below (a finite menu over two types), the feasible set is a compact subset of $\mathbb{R}^4$ cut out by linear IC/IR constraints and a continuous objective, so a maximizer again exists by Weierstrass's theorem; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability q^{k-m} with its first-order (small-risk) approximation around $q \rightarrow 1$: expected loss $\approx L(1 - q)(k - m)$, where L denotes the economic loss from a single unresolved-and-incorrect attribute. With $\Lambda \equiv L(1 - g)$, the user's payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2} m^2 - a c_Q (k - m).$$

4.2 The first-best benchmark

Before analyzing either regime, characterize the fully-informed, fully-aligned optimum: $(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m, a} U(m, a; \kappa)$.

Proposition 3 (First-best is a threshold rule in κ). *U is linear in a for fixed m , so $a^{FB}(\kappa) \in \{0, 1\}$. Define $\kappa^* = (\Lambda + c_Q)/(2k)$. If $\Lambda > c_Q$, then $a^{FB}(\kappa) = 1$ (always ask) for $\kappa > \kappa^*$ and $a^{FB}(\kappa) = 0$ (never ask, guess) for $\kappa < \kappa^*$.*

Proof. At $a = 1$: $m^*(\kappa; 1) = c_Q/\kappa$ and $U(m^*, 1; \kappa) = V - c_Q k + c_Q^2/(2\kappa)$. At $a = 0$: $m^*(\kappa; 0) = \Lambda/\kappa$ and $U(m^*, 0; \kappa) = V - \Lambda k + \Lambda^2/(2\kappa)$. Subtracting,

$$U(a=1) - U(a=0) = [\Lambda - c_Q] \left(k - \frac{\Lambda + c_Q}{2\kappa} \right),$$

which is positive iff $\Lambda > c_Q$ and $\kappa > (\Lambda + c_Q)/(2k) = \kappa^*$. $\square$

This is a clean, closed-form, type-dependent first best: high-cost (terse) users, who cannot cheaply self-provide detail, should always be asked; low-cost (verbose) users, who can cheaply write enough that the small residual is safe to guess, should never be asked. This threshold rule completely characterizes the efficient asking policy as a function of the user's marginal cost of detail relative to downstream error loss.

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ —it cannot condition on anything user-specific. This is the natural benchmark for a policy that is fixed at training time and cannot observe individual users.

5.1 User best response

The FOC $\partial U/\partial m = \Lambda(1 - a) - \kappa m + ac_Q = 0$ gives, with $\beta(\kappa) = \Lambda/\kappa$ and $\gamma(\kappa) = (\Lambda - c_Q)/\kappa$,

$$m^*(\kappa; a) = \beta(\kappa) - a\gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels $m^* \in (0, k)$; for low-cost types where unconstrained $\beta(\kappa) \geq k$, specification effort saturates at $m^* = k$. The closed-form population moments $\bar{R}_0$ and $\bar{\gamma}$ below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

Remark 3 (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of $\partial m^*/\partial a$ here does not depend on type: it is positive for every type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that holds, a higher population ask rate induces users to specify more, not less—they pre-empt costly clarifying questions by writing the answer down themselves. This is a genuinely new force relative to Section 2, where a only entered through accuracy, and is a distinct, directly testable prediction (P1' below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0, 1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.2 at the follower's best response $m^*(\kappa; a) = \beta(\kappa) - a\gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A U(m^*(\kappa; a), a; \kappa) - (\lambda_A - c_Q) a (k - m^*(\kappa; a)).$$

Substituting $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1-a)(k-m^*) - \frac{\kappa}{2}(m^*)^2 - a c_Q (k-m^*)$, using $k-m^*(\kappa; a) = R_0(\kappa) + a \gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A \left(V - \Lambda k + \frac{\Lambda^2}{2\kappa} \right) + (\mu_{AS} - b) R_0(\kappa) a + \frac{(\mu_{AS} - 2b) \gamma(\kappa)}{2} a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s \mathbb{E}[1/\kappa]$:

$$\Pi(a) = \text{const} + \bar{L} a + \bar{Q} a^2,$$

where $\text{const} = \mu_A(V - \Lambda k) + \frac{\mu_A \Lambda^2}{2} \mathbb{E}[1/\kappa]$, and

$$\bar{L} \equiv (\mu_{AS} - b) \bar{R}_0, \quad \bar{Q} \equiv \frac{(\mu_{AS} - 2b) \bar{\gamma}}{2}.$$

Proposition 4 (Stackelberg-optimal pooling rate). *Provided $\bar{Q} < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} = -\frac{\bar{L}}{2\bar{Q}} = -\frac{(\mu_{AS} - b) \bar{R}_0}{(\mu_{AS} - 2b) \bar{\gamma}}.$$

When this stationary point lies in $[0, 1]$, it is the unique global maximum of $\Pi(a)$ on $[0, 1]$.

Corollary 2 (Bias shifts the pooling rate on the interior branch). *Differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives, on the interior branch $\bar{Q} < 0$,*

$$\frac{\partial a^{SE}}{\partial \lambda_A} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_{AS}}{(\mu_{AS} - 2b)^2} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A(\Lambda - c_Q)}{[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)]^2},$$

which is strictly positive whenever $\bar{R}_0 < 0$ (the under-specification regime). Hence, a^{SE} is strictly increasing in perceived friction λ_A throughout the interior branch. This comparative static holds on the interior branch ($\bar{Q} < 0$); for the global optimum in the corner regime, see Corollary 4.

Corollary 3 (Non-identification of the two bias channels in the interior regime). *On the interior branch ($\bar{Q} < 0$), defining the normalized excess friction per unit of altruism $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A} = \frac{b}{\mu_A}$, the optimal pooling rate simplifies to*

$$a^{SE} = -\frac{\bar{R}_0}{\bar{\gamma}} \left[\frac{s - \tilde{\rho}}{s - 2\tilde{\rho}} \right],$$

so a^{SE} depends on the two bias parameters (μ_A, λ_A) strictly through the single scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$ (holding primitives s and $\bar{R}_0/\bar{\gamma}$ fixed). Consequently,

$$\frac{\partial a^{SE}/\partial \mu_A}{\partial a^{SE}/\partial \lambda_A} = -\frac{\lambda_A - c_Q}{\mu_A} = -\tilde{\rho},$$

and any two parameter pairs on the same ray $(\lambda_A - c_Q)/\mu_A = \text{const}$ produce the identical pooling ask rate. A leader that discounts downstream welfare ($\mu_A < 1$, $\lambda_A = c_Q \implies \tilde{\rho} = 0$) cannot be distinguished, from a^{SE} alone, from any leader along the corresponding ray of excess friction.

Remark 4 (Scope of Corollary 3). *The derivation above is stated on the interior branch $\bar{Q} < 0$. By Corollary 5 below, the unbiased pooling optimum is always a corner ($\bar{Q} \geq 0$), so the interior stationary point describes behavior when friction bias is sufficiently large ($2b > \mu_A s$).*

Corollary 4 (Ray-invariance extends to the corner regime). *On the corner branch ($\bar{Q} \geq 0$, or whenever the unconstrained stationary point lies outside $[0, 1]$), the optimal policy is chosen by the sign of the corner difference $\Pi(1) - \Pi(0) = \bar{L} + \bar{Q}$:*

$$\Pi(1) - \Pi(0) = \mu_A \left[s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) - \tilde{\rho} (\bar{R}_0 + \bar{\gamma}) \right].$$

Since $\mu_A > 0$, μ_A factors out completely, and the sign depends solely on the scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$. The critical threshold ratio is

$$\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}},$$

with $a^{SE} = 1$ for $\tilde{\rho} < \tilde{\rho}^*$ and $a^{SE} = 0$ for $\tilde{\rho} > \tilde{\rho}^*$. Ray-invariance in the excess friction ratio $\tilde{\rho}$ therefore holds unconditionally across the entire parameter space, interior and corner regimes alike. A numerical sweep of 11 rays across 10 values of $\mu_A \in [0.1, 1.0]$ (110 evaluations) confirms this analytical threshold with zero boundary-crossing violations.

Corollary 5 (Pooling is a corner solution in the unbiased case). *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\bar{Q} = \frac{s^2 \bar{\gamma}}{2} = \frac{(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa]}{2} \geq 0 \quad \text{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0, 1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0, 1\}$, not an interior compromise. Comparing the two corners directly:

$$\Pi(1) - \Pi(0) = \bar{L} + \bar{Q} = s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) = (\Lambda - c_Q) \left[k - \frac{\Lambda + c_Q}{2} \mathbb{E} \left[\frac{1}{\kappa} \right] \right],$$

so the unbiased pooling optimum is $a^{SE} = 1$ (always ask) when $k \geq \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$ and $a^{SE} = 0$ (never ask) when $k < \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$. This threshold matches the population-averaged first-best preference $\mathbb{E}[U(1) - U(0)]$ from Proposition 3 identically.

This last point matters for the comparison in Section 7: Model I has two independent, additive sources of loss relative to first best—the pooling restriction, and bias.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve is $\text{MRS}(m, a; \kappa) = -(\partial U/\partial m)/(\partial U/\partial a)$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q(k - m)$, we have $\partial U/\partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U/\partial a = s(k - m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\text{MRS}(m, a; \kappa) = \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \quad \frac{\partial \text{MRS}}{\partial \kappa} = \frac{m}{s(k - m)} > 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher- κ types therefore have strictly steeper indifference curves in (m, a) -space: they require more compensating ask-service per unit of specification effort given up. This satisfies the standard Spence–Mirrlees single-crossing condition (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 5 (Efficiency survives private information when the leader is unbiased). *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type’s own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$, $U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$ trivially: no bundle, including another type’s, can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader’s payoff from serving each true type is exactly maximized at that type’s own first best, so this menu is also leader-optimal. $\square$

This is the paper’s sharpest result: unlike classical regulation (Baron–Myerson, Laffont–Tirole), where the principal pays a literal transfer and therefore has an independent reason to dislike conceding rent, here the leader’s payoff can coincide exactly with user welfare, and when it does, incentive compatibility is *free*—private information costs nothing. Distortion in this setting is therefore a pure signature of misalignment, not of asymmetric information per se.

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader’s preferred bundle per type, $(m_\kappa^B, a_\kappa^B) = \arg \max_{m, a} \Pi_\kappa(m, a)$ (computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A -weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user’s true $U(\cdot; \kappa)$ now rank bundles differently.

Proposition 6 (Downward distortion of the high-cost type’s service). *Under the single-crossing condition and the empirically motivated under-asking bias ($\lambda_A > \mu_{ACQ}$, i.e. the leader over-weights the friction of asking relative to true user cost), the leader’s optimal incentive-compatible menu sets $a_L^{SB} = a_L^B$ (no further distortion for the low-cost type, whose biased first-best is already the binding corner $a = 0$) and $a_H^{SB} < a_H^B \leq a_H^{FB} = 1$ (the high-cost type is asked strictly less than the leader’s own already-biased preference for them), with a_H^{SB} determined by the binding incentive constraint IC_L .*

Proof. Form the Lagrangian of the leader’s screening problem relaxed to the binding upward-mimicking constraint IC_L :

$$\mathcal{L} = f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H) + \mu_{IC_L} \left[U(m_L, a_L; \kappa_L) - U(m_H, a_H; \kappa_L) \right],$$

with Lagrange multiplier $\mu_{IC_L} > 0$. Unlike classical transfer-based screening (Baron–Myerson, Laffont–Tirole), where the principal pays cash transfers to extract surplus—so the standard technique pushes the inefficient type’s participation constraint IR_H to reservation to minimize informational rent—here Π_{κ_H} already incorporates $\mu_A U(\cdot; \kappa_H)$ directly. The leader’s payoff from serving the high-cost type therefore *rises* with that type’s true welfare: there is no budget motive to depress U_H to $\underline{U}$, and IR_H remains strictly slack.

The low-cost type’s bundle (m_L, a_L) sits at its unconstrained biased optimum $(m_L^B, 0)$, as no other type seeks to mimic it under single-crossing. The high-cost type’s ask rate a_H^{SB} is chosen to maximize $\mathcal{L}$, satisfying the Kuhn–Tucker condition:

$$f_H \frac{\partial \Pi_{\kappa_H}}{\partial a_H} = \mu_{IC_L} \frac{\partial U(m_H, a_H; \kappa_L)}{\partial a_H} = \mu_{IC_L} s(k - m_H).$$

Because $\mu_{IC_L} > 0$ and $s(k - m_H) > 0$ for all $m_H < k$, the right-hand side is strictly positive. Consequently, $\partial \Pi_{\kappa_H} / \partial a_H > 0$ at the optimum, forcing a_H^{SB} strictly below the leader’s unconstrained biased maximizer a_H^B (downward distortion). $\square$

Remark 5 (Active constraint structure without monetary transfers). *In classical transfer-based screening (e.g., Baron–Myerson), IC_L and IR_H typically bind while IC_H and IR_L remain slack. In our setting, solving the full four-variable constrained problem shows that IC_L binds alone, with IR_H , IC_H , and IR_L remaining strictly slack at representative under-asking-bias parameters. This structural divergence arises because the leader’s payoff Π_{κ_H} directly internalizes follower welfare $\mu_A U(\cdot; \kappa_H)$, eliminating the classical principal’s motive to compress agent rent down to the reservation utility. We note that the exact active constraint configuration is parameter-region-specific: under sufficiently extreme asking friction ($\lambda_A \gg c_Q$) or severe discounting ($\mu_A \ll 1$), IC_H can bind alongside IC_L . The robust qualitative insight is that screening under misalignment strictly distorts $a_H^{SB} < a_H^B$, while the active constraint configuration reflects the non-monetary, partially aligned nature of the objective.*

Remark 6. *The direction in Proposition 6 is specific to the under-asking bias case ($\lambda_A > \mu_{ACQ}$), which we take to be the empirically relevant one (Section 3.3’s motivating story: turn-level satisfaction signals penalize extra clarifying questions more than users actually mind answering them). Under the opposite bias, the direction of distortion can flip; we do not claim a sign that holds for all Π_{κ} .*

7 Comparing the Two Regimes

Corollary 6 (Menus weakly dominate pooling). *The leader’s optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC , so the unconstrained menu optimum is at least as good.*

The more interesting comparison is *where the two regimes’ losses come from*, which Propositions 4–6 let us decompose cleanly:

- **Model I (pooling)** loses welfare from two distinct sources: (i) the pooling restriction itself, present even when $\mu_A=1, \lambda_A=c_Q$, where it manifests as a corner solution that fully serves one type and fully fails the other (Corollary 5), and (ii) bias, which shifts a^{SE} (Corollary 2) and can additionally shift pooling between corner and interior regimes. The two sources interact non-linearly across the parameter space.
- **Model II (screening)** loses welfare from bias *only* (Proposition 5): with no bias, screening recovers full efficiency despite private information. With bias, the loss is the rent-extraction-driven downward distortion of Proposition 6, which is a genuinely different mechanism from Model I’s pooling loss (it doesn’t arise from being unable to condition on anything—it arises from not being able to trust what is conditioned on).

7.1 Sensitivity of the Unbiased Pooling Corner

While Corollary 5’s qualitative finding—that unbiased pooling is always a corner solution rather than an interior compromise—is structural and universal, the specific corner selected depends on whether the task size k exceeds the population threshold $k^* \equiv \frac{\Lambda+c_Q}{2} \mathbb{E}[1/\kappa]$. Table 1 illustrates this boundary across varying task sizes k and baseline guessing accuracies g , holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

g	$\Lambda = L(1 - g)$	Threshold k^*	Task Size k	$\Pi(1) - \Pi(0)$	Optimal Corner a^{SE}
0.30	7.0	12.36	5	−36.80	0 (never ask)
0.30	7.0	12.36	10	−11.80	0 (never ask)
0.30	7.0	12.36	15	+13.20	1 (always ask)
0.50	5.0	9.61	5	−13.84	0 (never ask)
0.50	5.0	9.61	9	−1.84	0 (never ask)
0.50	5.0	9.61	10 (Benchmark)	+1.16	1 (always ask)
0.50	5.0	9.61	12	+7.16	1 (always ask)
0.50	5.0	9.61	15	+16.16	1 (always ask)
0.70	3.0	6.87	5	−1.87	0 (never ask)
0.70	3.0	6.87	8	+1.13	1 (always ask)
0.70	3.0	6.87	10	+3.13	1 (always ask)

Table 1: Sensitivity of the unbiased pooling corner choice $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ across task complexity k and guessing capability g , for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$. At the paper’s canonical benchmark ($g = 0.5, k = 10$), asking yields a modest advantage of +1.16, which flips to never-asking at $k \leq 9$.

Remark 7 (Compounding, testable). *Because Model II’s distortion is triggered by, and scales with, the same bias parameters (μ_A, λ_A vs. c_Q) that drive Model I’s shift in a^{SE} , reducing measured assistant bias should improve outcomes for high-cost (terse) users under both regimes, but the welfare gap between menu-conditioned and pooled policies should track population heterogeneity in κ (the spread $\kappa_H - \kappa_L$ and how far F sits on either side of κ^*) almost independently of bias. This gives two separable, testable quantities from the same field data: a bias-sensitive term and a heterogeneity-sensitive term.*

8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy should be associated with shorter specifications but higher or equal success rates—never both shorter specs and lower success.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length should flip sign depending on whether $c_Q \geq \Lambda = L(1 - g)$: where clarifying questions are cheap relative to the risk they resolve, users should specify less as ask rate rises (substitution); where questions are relatively costly to answer, users should specify *more* (pre-emption). This is a within-model prediction distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success should weaken, or invert, specifically for under-asking assistants, and can now be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users should shift specification length toward the sophisticated best response of Section 3.3.
- P4** (Prop. 3) Ask rates should rise with a user’s revealed difficulty specifying detail (terseness), sharply around a threshold κ^* , rather than responding only to model confidence.
- P5** Users with lower self-reported error-catching rates should show declining specification effort over a session (Section 3.2’s dynamic channel).
- P6** (Cor. 3) Under a pooling policy, terse users should be under-served and verbose users over-served relative to *each type’s own* first best, even for a demonstrably unbiased assistant—a gap attributable purely to instrument restriction, distinguishable in data from bias by holding λ_A fixed (e.g., via an A/B test on friction penalties in training) and varying only whether the policy can condition on declared specification.
- P7** (Prop. 6) Even where an assistant *can* condition on declared specification, terse users should still be asked less than the assistant’s own (possibly already-biased) per-type preference would predict, specifically because of the incentive-compatibility constraint—a distortion that should shrink as the population’s cost heterogeneity ($\kappa_H - \kappa_L$) shrinks, holding bias fixed (Remark on compounding).

9 Empirical Strategy (Summary)

The computational evaluation consists of closed-form analytical verification, exhaustive numerical grid sweeps, and the architectural design of LLM-based simulation experiments; no human-subject study is reported. Below we summarize the computational verification pipeline and outline a proposed human-study design for future empirical research.

Computational verification and simulation pipeline. The accompanying repository implements the simulation infrastructure for LLM-based user agents (instantiating simulated users with heterogeneous cost types κ interacting with assistants under Model I and Model II policies, with deterministic dry-run stubs for reproducibility). All quantitative findings and comparative statics reported in this paper are established via exact analytical derivations and exhaustive numerical grid sweeps across the parameter space. Specifically, we verify Proposition 3’s threshold

numerically, verify Proposition 4’s closed form against a fine grid search over a , and verify Proposition 5/6 by solving the full constrained menu optimization directly and checking that the low type’s bundle sits at its unconstrained optimum while the high type’s does not, with the gap shrinking as bias is set to zero.

Proposed human-subject study design. A between-subjects manipulation of ask policy (pooling vs. menu-conditioned; biased vs. unbiased friction weighting) across repeated coding tasks, measuring specification length and, where feasible, self-reported terseness/verbosity as a proxy for κ , to test P1’, P6, and P7 causally rather than correlationally.

10 Discussion and Limitations

The linear-risk approximation to the conjunctive success function is the main tractability assumption carried through Sections 4–7. To verify that this simplification does not drive our core theoretical findings, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ (independently of the linear-risk closed forms, via direct numerical grid search and constrained optimization). We find that the corner-solution property of Corollary 5 survives in every tested configuration across $k \in [3, 20]$ and $g \in [0.50, 0.95]$. Similarly, the downward-distortion direction of Proposition 6 survives across the parameter grid, with exceptions concentrated at small k where specification effort saturates at $m = k$ and the asking wedge collapses. Furthermore, a comprehensive parameter sweep of the full four-variable constrained menu problem under the exact conjunctive payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (100 grid points) confirms that IC_H binds alongside IC_L under extreme bias, reproducing the exact three-regime structure (no binding constraint, IC_L alone, $IC_L + IC_H$ together) identified under the linear-risk approximation, with only the boundary between moderate- and extreme-bias regimes shifting slightly (from $\lambda_A \approx 6.0$ to $\lambda_A \approx 8.0$ at $\mu_A = 1.0$). The extreme-bias reversal is therefore an intrinsic structural feature of the screening contract space, not an artifact of the linear-risk approximation.

A related limitation concerns identification of λ_A itself. Every bias-driven prediction (Corollary 2, Proposition 6, and predictions P2, P6, P7) is stated in terms of $\lambda_A > \mu_A c_Q$, but the paper does not currently propose a way to measure λ_A independent of the ask-rate behavior it is meant to explain; backing it out from observed policy alone is circular, and Corollary 3 sharpens this into an exact non-identification result on the interior branch: a^{SE} depends on (μ_A, λ_A) only through their ratio, so a discounting-driven and a friction-driven leader with the same λ_A/μ_A are indistinguishable from pooling behavior alone. An independent handle—for instance, from a training objective or reward specification that explicitly penalizes turn count—would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested non-circularly. The two-type restriction in Model II avoids ironing/continuum-type complications at the cost of generality; a continuum version using the standard Mirrlees technique is the natural next step and would let κ^* from the first-best benchmark interact directly with the screening distortion. We also assume truthful revelation of specified attributes (no cheap-talk vagueness within what is said) and a single leader (no competition among assistants, which would itself be an interesting extension—competition plausibly disciplines bias directly, another testable channel).

A broader methodological lesson from this model concerns the hazards of importing classical contract-theoretic intuition directly into human–AI settings. In classical mechanism design (e.g., Baron–Myerson, Laffont–Tirole), the principal pays monetary transfers to extract surplus, so agent rent directly diminishes principal payoff. In human–AI interaction, the assistant’s objective directly incorporates (a discounted multiple of) user welfare. As a result, incentive compatibility can be

free in the absence of bias (Proposition 5), participation constraints do not automatically bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 5). Contract-theoretic treatments of AI interaction must therefore explicitly account for the non-monetary, partially aligned nature of the interaction rather than relying on analogies to cash-transfer mechanisms.

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